[Kat]herine Titterton

San Francisco, CA · **macrofluidic.me** · X @KatTitterton

MACROFLUIDIC.ME/RESUME

TL;DR Interdisciplinary scientist-engineer at the seam of imaging, machine learning, and lab automation in biotech. 8+ years turning bench work into machine-learning fuel: building and scaling high-content assays, then the computational image analysis that unlocks phenotypic discovery at scale. Part process engineer, part artist-technologist who finds the shortest path and walks it.

EXPERIENCE

● **Peer-reviewed publication**

Founding Member of the Technical Staff, Axiom Bio · axi.om — *2023–Present · San Francisco, CA*

Founding technical hire building new approach methodologies (NAMs) for drug safety. Led feature engineering and algorithm development; oversaw data generation for two of the world's largest Cell Painting datasets (100,000+ compounds). Invented two bioactivity-detection methods ~10× more sensitive than conventional toxicity assays. Honorary Artist in Residence.

- Seal S, Dee W, Shah A, **Titterton K**, et al. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun* 2026.
- Ewald JD, **Titterton KL**, Bäuerle A, et al. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Syst* 2026.

Associate Scientist III, Insitro · Arrayed Screening, Neuroscience — *2022–2023 · South San Francisco, CA*

Adapted high-content phenotypic assays for high-throughput small-molecule screening (TSC, ALS) on near-fully-automated platforms (BioTek, High-Res). Built a digital phase-contrast timecourse imaging modality and Python analysis suite for label-free phenotypic discovery.

Technical Associate II, Massachusetts Institute of Technology · Synthetic Neurobiology Group — *2021–2022 · Cambridge, MA*

Optimized multiplexed expansion revealing (mExR) for 7+ rounds of IHC on 16× expanded specimens, enabling nanoscale visualization of 30+ proteins. Built image-analysis pipelines in ImageJ/Fiji and MATLAB.

- Kang J, Schroeder ME, Lee Y, **Titterton K**, et al. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nat Commun* 2024.

Associate Scientist II, AbbVie · Foundational Neuroscience Center, Early R&D — *2018–2021 · Cambridge, MA*

Built primary murine and hiPSC neuronal tau-aggregation high-content screening assays (Revvity Opera, Harmony). Modeled tau propagation in Alzheimer's disease with microfluidic devices; authored R dose-response pipelines.

- **Wu JW, Titterton K**, Sebastian R, et al. Development of a robust, physiologically relevant neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem* 2025.
- Sebastian RM, Kupp R, Romanul N, **Titterton K**, et al. SynCAR platform for capturing neuronal synaptic proteopathic seeds. *iScience* 2026.
- Manos JD, Preiss CN, Yanamandra K, **Titterton K**, et al. Uncovering specificity of endogenous TAU aggregation in a human iPSC-neuron TAU seeding model. *iScience* 2022.

Research Technician I, Massachusetts General Hospital — *2017–2018 · Boston, MA*

Used voltage-clamp electrophysiology in GABAR-expressing *Xenopus* oocytes to elucidate mechanisms of general anesthetics.

- Wu B, Jayakar SS, Zhou X, **Titterton K**, et al. Inhibitable photolabeling by neurosteroid diazirine analog in the β3-subunit of human type A GABA receptors. *Eur J Med Chem* 2019.

PROJECTS

Bare Metal — *2025*

First IRL art exhibit at Human Tech week, SF, Jun 2025. Post-hoc analysis on macrofluidic.me/blog.

Vivarium 2.0 — *2024*

A lavish habitat for Axiom Bio's pet poison dart frogs, rebuilt from scratch with automated misting and a motion-triggered camera that captures live video of the frogs.

OpenTronsformers (NYC) — *2023*

A companion automation engineer for wet-lab scientists: unlocking the OT-2 robot and Python SDK via LLM code generation. 2nd place at BioHack NYC 2023.

EDUCATION

Harvard University · Extension School, graduate coursework — *2017–2020*

Completed while employed full-time: Social Medicine; Intro to Programming (Python); Image Processing & Computer Vision (Python).

B.A., Colgate University · *cum laude* — *2013–2017*

Major: Cellular Neuroscience. Minor: Mathematical Systems Biology.

Coursework across systems biology, bioinformatics, biostatistics, and mathematical biology. Co-captain, Division I Track & Field; Dorothy-Waltham Award for Athletic Leadership.

SKILLS & TOOLS

COMPUTATIONAL

LLM codegen (Claude Code, Cursor) · image embeddings (visualization, clustering, feature derivation, normalization) · Python (data science, image processing & segmentation; Jupyter, NumPy, SciPy, scikit-learn, pandas, matplotlib, seaborn, xlsxwriter, Colab) · R (ggplot2, tidyverse, drc, platetools) · dose-response curve fitting (GraphPad Prism, R, Python) · uni/multivariate statistics · design of experiments (DOE) · ImageJ/Fiji macro scripting · Benchling ELN.

WET LAB

Cell culture (primary murine neuron, hiPSC neuron models) · microfluidic cell-culture models · ICC / IHC · confocal microscopy (manual, PE Operetta / Opera Phenix) · high-content image analysis (PerkinElmer Harmony) · automated liquid handling & schedulers (OpenTrons, Integra Assist Plus / Viaflo, BioTek, Labcyte Echo, High-Res Biosolutions, Green Button Go) · assay development & HT/HC screening · stereotaxic surgery / tissue microdissection · voltage-clamp electrophysiology.

CERTIFICATIONS

FII Level 1 Freediving (Oct 2019) · Molchanov L2 Freediving (Oct 2025) · SII Shark Ecology Specialty (Apr 2020)